

The entire air distribution system shall be balanced using an anemometer. Measured air quantities at fan discharge and at various outlets shall be identical to or less/excess than 5 percent in excess of those specified and quoted. Branch duct adjustments shall be permanently marked after air balancing is completed so that these can be restored to their correct position if disturbed at any time. Complete air balance report shall be submitted for scrutiny and approval, and four copies of the approved balance report shall be provided with completion documents.

INSULATION

SCOPE

The scope of this section comprises the supply and application of insulation conforming to these specifications.

MATERIAL

Thermal insulation material for Duct & Pipe insulation shall be closed cell Elastomeric Nitrile Rubber of Thermal conductivity of the insulation material shall not exceed 0.038 W/m²K or 0.212 BTU / (Hr-ft²-°F/inch) at an average temperature of 30°C. Density of the nitrile rubber shall be 40-60 Kg/m³. The product shall have temperature range of -40 °C to 105°C. The insulation material shall be fire rated for Class 0 as per BS 476 Part 6 : 1989 for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test. Water vapour permeability shall be not less than 0.024 per inch (2.48 x 10⁻¹⁴ Kg/m.s.Pa i.e. $\mu \geq 7000$: Water vapour diffusion resistance).

Comply

In addition to above properties, for Hospital, insulation shall be anti-microbial with EPA approval. Microbiological growth on insulation surface shall be in accordance with UL 181 and bacterial resistance to ASTM G22.

Thickness of the insulation shall be as specified for the individual application. **Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for thermal conductivity values, density, water vapour permeability and fire properties.** Samples of insulation material from each lot delivered at site may be selected by Owner's site representative and gotten tested for thermal conductivity and density at Contractor's cost. Adhesive used for sealing the insulation shall be non-flammable, vapour proof adhesive strictly as per manufacturer's recommendations.

Ducting insulation thickness shall be as per below:

FOR G.I. DUCTING-

THERMAL INSULATION-WITH NITRILE RUBBER

The supply air duct shall be insulated with 19/15/10 mm thick, Nitrile Rubber
Method of applying insulation-

- Clean the duct surface to be insulated.
- Apply a thin layer of tar paints.
- Apply a thin coat of rubber solution to stick the insulation.
- Fix the insulation of specified thickness over the surface of the duct tightly and seal all the joints

Noted



using BOPP tape. Secure the insulation with 16 Gauge G. I. wire or 10 mm wire PVC box strapping at a distance of 300 mm.

THERMAL INSULATION-

The supply air duct shall be insulated with 50/40/25 mm thick, factory backed Aluminium foil fiber-glass of density 24 Kg. / Cu. M.

Method of applying insulation-

Noted

- a) Clean the duct surface to be insulated.
- b) Apply a thin layer of tar paints.
- c) Apply a thin coat of hot bitumen or coat of bituminous paint to stick the insulation.
- d) Fix the insulation of specified thickness over the surface of the duct tightly and seal all the joints using BOPP tape. Secure the insulation with 16 Gauge G. I. wire or 10 mm wire PVC box strapping at a distance of 300 mm.

THERMAL INSUALTION-WITH CLOSED CELL EPE

The supply air duct shall be insulated with 25/19/13 mm thick, Closed Cell Expanded Polyethylene.

Method of applying insulation-

Noted

- a) Clean the duct surface to be insulated.
- b) Apply a thin layer of tar paints.
- c) Apply a thin coat of rubber solution to stick the insulation.
- d) Fix the insulation of specified thickness over the surface of the duct tightly and seal all the joints using BOPP tape. Secure the insulation with 16 Gauge G. I. wire or 10 mm wire PVC box strapping at a distance of 300 mm.

ACOUSTIC INSULATION-

First 3 meter length of supply air duct shall be acoustically insulated with 12.5 mm thick fiberglass of density 48 Kg./Cu.M. and covered with 28 G perforated Aluminium sheets from the inside of the duct.

Noted

- a) Apply a thin layer of tar paints.
- b) Fix-up fiberglass slabs
- c) Cover-up with perforated Aluminium sheets with the help of G. I. Screw Washers.

FALSE CEILING INSULATION-

The false ceiling shall be insulated with 50 mm thick fiberglass slab of 24 Kg. / Cu. M. density. The fiberglass slab shall be wrapped in polyethylene bags.

UNDERDECK, AHU ROOM AND SHAFT INSUALTION-

The exposed roof shall be insulated with 50 mm thick "TF" quality expanded polystyrene of density



16 Kg./Cu. M. with 85/20 grade hot bitumen and GI screw washer & GI diagonal wires. The insulation shall be covered with 36 G Aluminium Foil.

PVC DUCTING :

The ducting shall be of RIGID PVC of 2.5/4.0 kg/cm² pressure rating and suitable for high duct velocity. The duct shall be supported with 22 G G.I. sleeves, 10mm dia M.S. rod, Anchor fasteners etc. All the fittings viz. Elbows, bends, couplings, flanges, stubs, Reducers, etc. shall be joined by means of heat treatment to the fittings. However, the joints shall also be further sealed with the help of Binding solution for Extra protection. All outlet points shall be provided with volume control PVC damper / PVC Butterfly valves. On every

Noted

OVERDECK INSULATION

Overdeck insulation shall be done with 75 mm thick extruded polystyrene of density 45-48 kg/cm³ & thermal conductivity of 0.21 Btu in / ft² hr°F (at 24°C as per ASTM C – 518). Minimum compressive strength as per ASTM D-2842 shall be 570 kPa water absorption as per ASTM D-2842 shall not be more than 1%.

Method of Application

- (a) Clean RCC slab and make it free from dust and other laitance matter.
- (b) Lay cement based water proofing on roof with a minimum slope of 1:100 and average thickness of 110 mm using brickbats of appropriate size and shape suitable to achieve the required slope laid over 15 mm thick waterproof element mortar 1:4 and finished with 20 mm thick waterproof plaster wing cement mortar 1:4 and making false squares of 300 mm size including rounding off the junction of roof and parapet walls for a height of 300 mm with brickbats and 20 mm thick waterproof plaster and conducting necessary leakage / dampers tests, etc.
- (c) Lay 75 mm thick extruded polystyrene boards over prepared surface fixing with adhesive. Adhesive shall be strictly as per recommendations from manufacturer.
- (d) Lay 80 gsm geotextile fabric over insulation board
- (e) Lay 40 x 40 x 4 cm precast paver blocks.

Noted

(Note : If contractor is awarded work of waterproofing + over deck insulation, follow all steps from (a) to (e). if contractor is awarded work of only over deck insulation follow step (c) and (d). Rest will be done by Civil Contractor).

UNDERDECK INSULATION

Underdeck insulation shall be 50mm thick TF Quality expanded polystyrene (32 Kg/m³) or 30mm thick phenotherm. Underdeck surface of ceiling shall be cleaned and made dirt free. Insulation panels shall be pasted on this surface with black CPRX compound. 28g wire net shall be tightened around insulation so as to avoid any kind of sagging. Ends of net shall be overlapping by at least 25mm. Overlaps shall be screwed with galvanised screws to avoid rusting.



MEASUREMENT OF INSULATION

Unless otherwise specified measurement for duct and pipe insulation for the project shall be on the basis of centre line measurements described herewith

- a. Pipe Insulation shall be measured in units of length along the centre line of the installed pipe, strictly on the same basis as the piping measurements described earlier. The linear measurements shall be taken before the application of the insulation. It may be noted that for piping measurement, all valves, orifice plates and strainers are not separately measurable by their number and size. It is to be clearly understood that for the insulation measurements, all these accessories including 26G aluminium cladding, valves, orifice plates and strainers shall be considered strictly by linear measurements along the centre line of pipes and no special rate shall be applicable for insulation of any accessories, fixtures or fittings whatsoever.
- b. Duct Insulation and Acoustic Lining shall be measured on the basis of surface area along the centre line of insulation thickness. Thus the surface area of externally thermally insulated or acoustically lined be based on the perimeter comprising centre line (of thickness of insulation) width and depth of the cross section of insulated or lined duct, multiplied by the centre-line length including tapered pieces, bends, tees, branches, etc. as measured for bare ducting.

QUALITY ASSURANCE

The insulation material for the respective application shall be manufactured as per the International Standard.

The density and other material composition for correct quality shall be tested in the factory as per the International Norms and Codes.

OEMs QAP to indicate the quality check at manufacturers work.

All the contractors to do WPQ as per ASME section 9 prior to start of chilled water piping works.

Dye penetration test to be done in the any selected chilled water pipeline joints in presence of client representatives.

SUBMITTALS

BIDDING STAGE

1. Manufacturing data sheet conforming quality, K value thickness etc.
2. Certificate of fire classification obtained from an approved in testing lab.
3. Manufacturer test certificate.
4. Product data of material used for finishing to be provided over insulation material.

POST BIDDING

1. Test certificate of the insulation material to be used for piping, ducting , thermal, acoustic and false ceiling insulation.
2. Layout drawings indicating size, type and location of insulation used for the above application.
3. Details of adhesive material.
4. Procedure for carrying out insulation work for the respective application.
- 5 Insulation thickness calculation by both temperature rise and condensation control method to

Noted

